Bt501 Health Biotechnology
Final Term Notes (Past+ Current)
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CURRENT FALL 2023

Q: pharmaceutical technology

Pharmaceutical Technology

The methods and techniques that involve the use of living organisms (such as cells, bacteria, yeast and others) are tools to perform specific industrial or manufacturing process are called biotechnology. Pharmaceutical Technology will continue to provide new breakthroughs in medical research in the years to come, leading to treatment in field which have previously eluded us (including AIDS, cancer asthma, Parkinson‘s disease, Alzheimer disease)

Q: 5 effective ways to deliver vaccine

as early as 1994, researchers identified five effective ways for delivering influenza virus plasmids to the animal body: by nasal spray; by injection into the muscle, vein, and under the skin; and by a so-called gene gun

Q: seamless cloning

Seamless cloning technologies eliminate the requirement for restriction enzymes. This can be advantageous when an insert contains a number of restriction sites within its sequence, making it difficult to identify restriction enzymes that will not cut the gene of interest during the cloning procedure.

Benefits:

- Many copies of genetic material available in the laboratory for the purpose of experimentation

- Production of pharmaceutical drugs, insulin, clotting factors, human growth hormone, and industrial enzymes
- Important for human research
- For production of GMOs

Q: application of cDNA in medicine agriculture and pharmacology

Q: drug addiction

Drug Addiction: Chronic, often relapsing brain disease that causes compulsive drug seeking and use, despite harmful consequences to the addicted individual and to those around drug addicts.

Q: role of modern. And classical biotechnology

Classical Biotech vs Modern

- Classical Biotech –since ages ,Growing better crops, Breeding animals
- Cheese and yogurt production – use of microorganisms
- Mendel‘s work – Laying foundation of modern Biotech
- Major discoveries : DNA structure, Ligase, Restriction enzyme
- Birth of rDNA technology
- Cut Gene from one organism and paste into others

Q: define alpha hemolysine

ALPHA-HEMOLYSIN: a-Hemolysin (a-HL, also called a- toxin). a-HL is an exotoxin secreted by the bacterium Staphylococcus aureus, a human pathogen.This mushroom-shaped heptameric protein normally forms pores in the membranes.The limited pore size (1.4 nm) has restricted its application in the analysis of ssDNA, RNA, or small molecules.

Q: procedure or steps of monoclonal antibodies.

Monoclonal antibodies

Another milestone achieved in the beginning of the twentieth century was the making of monoclonal antibodies, which was proposed by Paul Ehrlich. He suggested that drug compounds

can be accurately, precisely, and specifically delivered along with monoclonal antibodies. Monoclonal antibodies are mono-specific antibodies as they are made by identical cells that are all clones of a distinctive parent cell. Currently, it is possible to produce monoclonal antibodies that specifically bind to that antigen and have a variety of applications, especially to detect or purify that substance (antigen), and has become an important tool in diagnostics and biomedical research. In diagnostics and biomedical research, monoclonal antibodies are very useful tools. First, they are extremely specific; that is, each antibody binds to the specific site of the antigen. Second, some antibodies, once activated by the occurrence of a disease, continue to confer resistance against that disease; classic examples are the antibodies to the childhood diseases chickenpox and measles. Another important application of monoclonal antibodies is to develop vaccines against various diseases. As you are aware, a vaccine is made from bacteria or viruses either killed or inactivated. Upon introduction into the human body this vaccine stimulates the production of antibodies against the antigens to fight back the diseases. The production of monoclonal antibodies involves human and mouse hybrid cells and this technology is known as hybridoma technology. During monoclonal antibody production, tumor cells are merged with mammalian cells that produce an antibody against a particular antigen. The result of these merged tumors with mammalian cells is a called hybridoma, which can frequently produce antibodies. These antibodies are called monoclonal antibodies because they come from only one type of cell; whereas, the antibodies that are produced by conventional methods contain many kinds of cells and are called polyclonal antibodies.

Cardiovascular disease symptoms, treatment

Disease	Possible causes	Affects	Symptoms	Treatment
Cardiovascular disease	Inherited and acquired, predominant lifestyle disease caused by smoking	Coronary arteries are at risk for narrowing as cholesterol deposits, called plaques, build up inside the artery, affect the heart muscle or the blood vessels of the heart which can lead to congestive heart failure, and may also cause disturbances in normal heartbeat, called arrhythmia; clot in artery due to rupture of plaque can lead to myocardial infarction	Chest pain, shortness of breath, sweating, nausea, and indigestion	Medical intervention with multistep approach, lifestyle modification, regular exercise, cholesterol reduction, quitting smoking and alcohol consumption, and stress-free life
	High blood pressure (hypertension)			
	High cholesterol			
	Diabetes			
	Family history			
	Peripheral artery disease			
	Obesity			

Recombinant insulin

Another discovery which was made in the medical field was the recombinant DNA technology, which has completely revolutionized disease treatments especially in the Type 1 diabetes mellitus, where insulin-producing cells become dysfunctional and do not produce sufficient amount of insulin to regulate blood-sugar level. In a healthy human being, insulin regulates glucose metabolism in the body. In diabetic conditions, the level of insulin decreases which causes the elevation of the blood sugar level and this clinical condition is known as diabetes mellitus. One of the best treatments of diabetes mellitus is insulin injections, where insulin is injected into the patient‘s body and the insulin regulates the bloodsugar level. The insulin is either synthesized chemically or produced by recombinant DNA technology. These synthesized insulins are used medically to treat patients with Type 1diabetes mellitus, whereas patients with Type 2 diabetes mellitus are insulin resistant.

Difference between Drug abuse or drug addiction

Drug Abuse: A form of substance-related disorder. Patterned use of a drug in which the user consumes the substance in amounts or with methods which are harmful to themselves or others. Drugs most often associated with this term include: Alcohol, substituted amphetamines, barbiturates, benzodiazepines, cocaine, methaqualone, and opioids.

Drug Addiction: Chronic, often relapsing brain disease that causes compulsive drug seeking and use, despite harmful consequences to the addicted individual and to those around drug addicts.

Difference between drug abuse and addiction

- Drug abuse refers to using drugs even though it has become a problem in life.
- Drug addiction refers to the substance, taken in higher amounts or taken over a longer period of time than the user intended.

pharmacogenomics

Pharmacogenomics: is the science of how genes affect the way people respond to drugs

- How genes affect, the way our body processes drugs (pharmacokinetics)
- The interaction of drugs with receptors (pharmacodynamics)

- The treatment efficacy and adverse side effects
- A subset of _pharmacogenomics‘
- The study of how inherited variation affects drug response and metabolism

What is vaccine and types and define each type? 3

VACCINE: The vaccine is basically a biological preparation that improves the human immune defense system to fight diseases. The vaccines normally contain an agent that diligently resembles a disease-causing bacteria, and is frequently produced from debilitating or killed forms of the microorganisms

Type of vaccines

Vaccines are produced by either dead or inactivated organisms and there are four main ways to develop vaccines:

(i) live attenuated vaccines contain bacteria or viruses that have been altered so they cannot cause disease,

(ii) killed vaccines contain killed bacteria or inactivated viruses

(iii) toxoid vaccines contain toxins (or poisons) produced by the germ that have been made harmless

(iv) component vaccines contain parts of the whole bacteria or viruses.

What is erythropoetin? 3

Like insulin, erythropoietin (EPO) is a glycoprotein hormone that controls red blood cell production in the body (Figure 1). EPO, also called hemopoietin, is known to be located as precursors in the bone marrow. It is produced by the peritubular capillary endothelial cells in the kidney and liver. Besides its role in red blood cell production, EPO also plays an important role in neuronal injury and in the wound healing process. It has been reported that EPO has its primary effect on red blood cells by protecting the blood cells from going to the apoptosis pathway. It also conjoins with various factors involved in the growth of precursor red cells. EPO has a range of actions in hypertension, stimulating angiogenesis, and inducing proliferation of

smooth muscle fibers. It has also been shown that EPO can upsurge iron absorption by quashing the hormone hepcidin.

What is transcriptome? 3

Transcriptome: The transcriptome is the complete set of transcripts in a cell and their quantity, for a specific developmental stage or physiological condition. Understanding the transcriptome is essential for interpreting the functional elements of the genome and revealing the molecular constituents of cells and tissues, and also for understanding development and disease.

What are innate and outer immune system? 3

Two major types of immunity are carried out by cells of the immune system in response to inflammation and infection (Fig.1).

- First, innate immunity provides a mechanism of host defense that mediates the initial protection against microbial infections.
- Second, adaptive immunity, which develops more slowly and confers specificity against a foreign antigen, mediates the later and more vigorous defense against infections.

What are techniques of immune system? 5

Four important functions are performed by the immune system:

1. First, immunological recognition is conducted by leukocytes (e.g., neutrophils, macrophages, and natural killer cells) of the innate immune system that provide an early rapid response and by the lymphocytes (T cells and B cells) of the later, more antigen-specific and more efficient adaptive immune system.
2. Second, to contain and eliminate the infection, immune effector functions that involve the complement system of blood proteins, enzymes, and antibodies together with certain T cells and B cells are required.
3. Third, while combating foreign pathogens, the immune system must learn not to elicit damage to self-tissues and other components of the body. It accomplishes this by immune regulation, a process in which regulatory lymphocytes control various immune responses against self-components. A failure in immune regulation could potentially result in allergy and autoimmune disease.
4. Fourth, the immune system must be equipped to protect an individual against recurring disease upon exposure to the same or a closely related pathogen. This task is uniquely

mediated by the adaptive immune system during a response known as immunological memory. After exposure to an infectious pathogen, a person will make a rapid and more vigorous response to the pathogen upon subsequent encounter, with this pathogen and will acquire protective immunity against it. Thus, a very significant challenge facing immunologists today is the identification of relevant cellular pathways and molecule that generate long-lasting immunity (i.e., memory) to pathogens, with the aim of identifying novel immunotherapies using cells and/or antibodies that eradicate such pathogens.

What is blood typing and cross matching? 5

Typing and Cross Matching

Before any blood can be given to a patient, it must be determined that the blood of the donor is compatible with the patient. The laboratory examination to determine a person‘s blood group and Rh factor is called Blood Typing. The process of determining compatibility between blood specimens is Cross matching. RBCs from the donor blood are mixed with serum from the recipient, a reagent (Coombs‘serum) is added, and the mixture is examined for visible agglutination. If no antibodies to the donated RBCs are present in the recipient‘s serum, agglutination does not occur and the risk of transfusion reaction is small.

Ethical recommendations to resolve issues of gene therapy? 5

Recommendation to resolve the Legal and ethical issues

Nevertheless, there are some ethical concerns which have been raised in different parts of the world, and some ethical organizations have released recommendations to resolve the gene therapy issues and these recommendations include:

(a) establishing a national ethics body in each country to look at somatic gene therapy

(b) supporting somatic gene therapy research that follows the recommendations

(c) asking researchers, organizations, and governments to listen and respond to public concerns about gene therapy research

(d) asking that research follows quality and safety controls.

YEAR 2020

Qno1: Define Mitochondrial Homeostasis.

Because neurons require considerable energy to maintain their metabolically active state, they rely heavily on the maintenance of mitochondria in a functional state, a property known as mitochondrial homeostasis.

Qno2: Write down four important function of Immune system.

1. Immune reorganization

2. Contain and eliminate infection

3. Immune regulation

4. Immunological memory

Qno3: Write down the modifications which are also there for creating transgenic.

A few modifications are also there for creating transgenic:

- Electroporation of DNA into the sperm, ova, or embryos.
- Nuclear transfer with somatic or embryonic cells.

Qno4: Why CA125 test is not recommended as a screening test of ovarian cancers?

CA125 test is not recommended as a screening test of ovarian cancers, because CA125-a high molecular-weight glycoprotein

- Tumor Marker Non specific –increase in other tumors/noncancerous disorders Measuring Disease-Associated
- Proteins by Sandwich ELISA
- Research- other biomarkers -more specific for ovarian cancer.

Qno5: Define passive immunity.

Development of immunity between the mother and child is called passive immunity because the fetus does not make any antibodies on its own; it only borrows them from the mother.

Qno6: Write about the somatic cell nuclear transfer technique.

- Due to the absence of proven ES cells and the recent advances in nuclear transfer (NT), current emphasis for creating transgenic animals is on somatic cell nuclear transfer (SCNT).
- Nuclear transfer is a technique that is used to create a genetically identical copy, or a clone, of an animal.
- Nuclear transfer commonly involves the transfer or placement of a donor nucleus into the cytoplasm of an enucleated MII oocyte.

Qno7: Define Autologous cells.

The autologous cell transplantation is basically a transplantation in which cells obtained from patients can be transplanted into the same patient.

Qno8: Write down two advantages of Live attenuated vaccine.

The advantages of attenuated vaccines Single dose is effective, and boosters are not

- Essentially required because immune system is exposed to the live organism for prolonged period.
- Efficiently activates adaptive immune responses.

3 Marks

Qno1: Write about the Hemagglutination test to determine blood type.

Hemagglutination test to determine blood type.

(A) Antiserum containing antibodies against A or B surface antigens is mixed with red blood cells carrying A

surface antigens. Clumping (agglutination) in the presence of anti-A antiserum but not with anti-B antiserum indicates blood type A.

(B) Agglutination responses for different blood types.

Qno2: How many types of immune disease. Write the name.

There are three types of immune disease.

a) Immunodeficiency

b) Autoimmune disease

c) Hypersensitivity

Qno3: Write down a note on Recombinant blood clotting factor VIII.

Factor VIII is one of the important factors of all blood clotting factors. Deficiency of factor VIII causes bleeding disorder called hemophilia A. The hemophilia may be mild to severe depending upon factor VIII concentration in the body. In moderate and severe factor VIII deficiency, there can be spontaneous bleeding episodes in the joints. Hemophilia A affects 1 in 5,000–10,000 males. Replacement therapy is the treatment option for hemophiliacs either with human plasma-derived factor VIII (pdFVIII) or recombinant FVIII (rFVIII). Transfection of HEK 293 cell cultures in serum-free suspension is being tried for optimal yield. Recombinant factor VIII (rFVIII) is produced by culturing mammalian cells as baby hamster kidney (BHK) or Chinese hamster ovary cells (CHO), using large-scale bioreactors . Standardizations are done to maximize yields.

Qno4: Write the three trisomy with disorder name and number of chromosome.

- Trisomy 13, the presence of three copies of chromosome 13, causes Patau syndrome.
- Trisomy 18, the presence of 3 copies of chromosome 18, elicits Edward syndrome.
- Trisomy 21, the first human chromosomal disorder discovered (in 1959), is an abnormality that displays an extra copy (total of 3 copies) of chromosome 21 and causes Down syndrome.

Qno5: Define immunodeficiency and its types.

Immunodeficiency is a physiological condition where the immune system’s capacity to fight infectious disease has become dysfunctional or entirely absent.

Primary immunodeficiency

A number of rare diseases have been reported susceptible to infections from childhood onward and many of these diseases are hereditary and autosomal recessive or X-linked in nature. There are more than 80 primary immunodeficiency syndromes reported in humans and they are generally grouped by their malfunctioning of lymphocytes or granulocyte cells. The treatment of primary immunodeficiencies depends on the nature of the defect, and may involve antibody infusions, long-term antibiotics, and stem cell transplantation.

Acquired immunodeficiency

An immune deficiency that is caused due to external processes is called secondary or acquired immunodeficiency. The common causes for secondary immunodeficiency are malnutrition, aging, and particular medications. It has been reported that there are many specific diseases that directly or indirectly impair the immune system Immunodeficiency is also the hallmark of AIDS, caused by HIV.

Qno6: Write down two basic strategies for producing transgenic animals.

There are two basic strategies for producing transgenic animals.

• These are the so-called "gain of function" or "loss of function" transgenics.

• The "gain of function" relies on addition of a cloned fragment of DNA to an animal's genome to obtain new expression of a gene product that did not previously exist in that cell or tissue type.

Qno7: Write hybridization probe which one of DNA sequence is specific diagnostic approach.

Hybridization is the formation of hydrogen bonds between two complementary strands of nucleic acids. A diagnostic test involving DNA hybridization utilizes a DNA probe to detect a complementary target DNA sequence that is characteristic of the disease. The probe is labeled with a reporter molecule that indicates hybridization between the target and probe DNA. Hybridization probes are often employed to detect the presence of microbial pathogens. Malaria, caused by the parasite Plasmodium falciparum, is one of the most common infectious diseases and is especially fatal in young children.

Qno8: Define and classify the immunological memory.

When B cells and T cells are activated and begin to reproduce, some of their offspring can become long lived memory cells. This process is called immunological memory. Immunological memory can be classified as passive short-term memory or active long-term memory.

5 Marks

Qno1: Write a note on Fragile X syndrome.

Fragile X syndrome is caused by a ―fragile‖ site at the end of the long arm of the X chromosome. Fragile X syndrome is the most common cause of inherited mental retardation.

Although it is an X-chromosome-linked recessive trait with variable expression and incomplete penetrance, 30% of all carrier women are affected. Fragile X syndrome is caused by loss-of-function mutations in the fragile X mental retardation 1 (FMR1) gene.

FMR1 encodes the FMRP protein found in many tissues and at particularly high levels in the brain and testes. The FMRP protein may help regulate synaptic plasticity and thereby control learning and memory. Fragile X syndrome belongs to a growing class of neurodegenerative disorders known as trinucleotide repeat disorders. Among these disorders, 14 affect humans and elicit neurological dysfunction. Trinucleotide CGG repeat expansions (200 to more than 1,000 repeats) that inactivate the FMR1 gene are the most common mutations observed at this locus.

Qno2: Write down the advantages of PCR based test.

Advantages: specificity, sensitivity, amplification (subnational amount of a target sequence), multiplexing, low cost.

Qno3: Write down the advantages of DNA Vaccine.

The advantages:

• Protein is expressed in its native conformation.

• Long-term exposure of immune system to the antigen.

• Efficient activation of cell-mediated and humoral immune systems leading to memory response.

• There is no requirement of specific storage condition like refrigeration of plasmid DNA, and it reduces the cost and is easy to handle.

• The genes expressing cytokines can be ligated to the vector, and the cytokine produced at the site may activate the desirable branch of immune system.

• Vaccine is safe, without any infectious agent.

• Easier to produce in large quantity.

Qno4: Write a note on Genetic Tay Sach disease.

Tay–Sachs disease is an autosomal recessive fatal genetic disorder caused by a genetic mutation in the HEXA gene on human chromosome 15, which encodes the hexosaminidase A enzyme.

This mutation leads to a decrease in function of hexosaminidase A, and as a result, harmful quantities of a sphingoglycolipid, termed ganglioside GM2, accumulate in brain neurons.

In the most common form of the disease (infantile Tay–Sachs), abnormal hexosaminidase A enzyme activity and the accompanying harmful accumulation of cell membrane associated gangliosides in neurons lead to premature neuronal death, paralysis, dementia, blindness, psychoses, and even death of the patient. Tay–Sachs disease may be diagnosed by a blood test that measures levels of the hexosaminidase A enzyme in serum, lymphocytes, or skin fibroblasts. Prenatal tests of hexosaminidase A activity, such as amniocentesis (at 15 to 16 weeks of pregnancy) and chorionic villus sampling (at 10 to 12 weeks of pregnancy), can now diagnose Tay–Sachs disease in the fetus.

Qno5: Write a note Recombinant Human Insulin.

Insulin is a peptide hormone consisting of 51 amino acids. It is secreted by β cells of islets of Langerhans of pancreas. The hormone is responsible for maintaining normal blood glucose level in blood. Insulin is stored in the form of proinsulin which contains two polypeptide chains, A and B, and is connected with a third peptide C chain, which before secretion is cleaved with production of insulin and C-chain. The cleavage results in the removal of C-chain, and the A (21 amino acids) and B chain (30 amino acids) are linked by disulfide linkage to form mature insulin

In the beginning, efforts were made to isolate mRNA for pre- and proinsulin from rat islets of Langerhans of pancreas and to synthesize cDNA. Thereafter, it was inserted into a plasmid. The recombinant plasmids were transferred into the E. coli cells, which secreted proinsulin. Scientists have chemically synthesized DNA sequences for two chains, A and B, of insulin and separately inserted into two pBR322 plasmids by the side of β-galactosidase gene.

Qno6: Define rDNA technology and write its step of gene cloning in rDNA technology.

Transgenic organisms are genetically modified organisms (GMOs) which have inserted DNA of different species origin. The inserted DNA/gene is known as transgene. Transgene is introduced into the organism, using recombinant DNA technology.

Gene cloning has four stages:

(1) Generation of DNA fragments,

(2) Ligation to a vector,

(3) Propagation in a host cell,

(4) Selection of the transformed host.

Qno7: How can two different alleles are distinguish by using a Taq man Assay.

To simultaneously detect normal and disease-associated alleles, two TaqMan probes are utilized. Each probe is exactly complementary to either the normal or the disease-associated DNA sequence, and each probe has a different fluorescent dye attached to its 5′ end (Fig.). Intact probes, whether unbound or bound to template DNA, do not fluoresce because of the presence of a quencher molecule at the 3′ end of the probe. PCR primers anneal to sequences that flank the probe hybridization site, and as PCR amplification proceeds, the TaqMan probe is displaced by the growing DNA strand. The 5′ nuclease activity of the Taq polymerase degrades the 5′ end of the TaqMan probe, thereby releasing the fluorescent dye and removing it from the proximity of the quencher molecule. Thus, only TaqMan probes that were previously bound to target DNA are degraded and subsequently fluoresce. Any mismatched probes that are not complementary to the SNP in the region where the TaqMan probe binds will be displaced but not cleaved, so they will not fluoresce. By monitoring the fluorescence at two different wavelengths (one for each TaqMan probe), it is possible to distinguish individuals that are homozygous for each SNP (either normal or disease associated) and heterozygous (carrying one normal and one disease-associated allele). This technique may be used to detect more than two different SNPs at the same time as long as the fluorescent dyes attached to each SNP-specific probe have well-separated, non-overlapping fluorescence maxima.

Write a note on embryonic stem cell mediated gene-transfer method for producing transgenic animals.

Embryonic stem cells are obtained from inner cell mass of blastocyst. This method involves prior insertion of the desired DNA sequence by homologous recombination into an in vitro culture of embryonic stem (ES) cells, and the transformed cells can be selected using standard markers. Cells containing the desired DNA are incorporated into the blastocoel of host's embryo at the blastocyst stage, where they mix with inner cell mass. Unlike the other two methods, which require live transgenic offspring to test for the presence of the desired transgene.

This method allows testing for transgene at the cell stage. This is advantageous as there is no convenient way to select for eggs or embryos that have taken up foreign DNA. ES cell-mediated gene transfer is the method of choice for gene inactivation, the so-called knockout method. This technique is of particular importance for the study of the genetic control of developmental

processes. This technique works particularly well in mice. It has the advantage of allowing precise targeting of defined mutations in the gene via homologous recombination.

Define rDNA technology and write its step of gene cloning in rDNA technology.

V. Important

Transgenic organisms are genetically modified organisms (GMOs) which have inserted DNA of different species origin. The inserted DNA/gene is known as transgene. Transgene is introduced into the organism, using recombinant DNA technology.

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(2) Ligation to a vector,

(3) Propagation in a host cell,

(4) Selection of the transformed host.

There are two basic strategies for producing transgenic animals.

• These are the so-called "gain of function" or "loss of function" transgenics.

• The "gain of function" relies on addition of a cloned fragment of DNA to an animal's genome to obtain new expression of a gene product that did not previously exist in that cell or tissue type.

YEAR 2021

Q1. Somatic cell nuclear transfer. 2 Marks

Due to the absence of proven ES cells and the recent advances in nuclear transfer (NT), current emphasis for creating transgenic animals is on somatic cell nuclear transfer (SCNT). Nuclear transfer is a technique that is used to create a genetically identical copy, or a clone, of an animal. Nuclear transfer commonly involves the transfer or placement of a donor nucleus into the cytoplasm of an enucleated MII oocyte.

Q2. Who was the First test tube baby? 2 Marks

Birth of Louise Brown in 1978, world's first test tube baby.

Q3. Immunological memory and its types? 3 Marks

When B cells and T cells are activated and begin to reproduce, some of their offspring can become long-lived memory cells. This process is called immunological memory. Memory cells can remember each and every specific pathogen encountered and can enhance a strong response if the pathogen is detected or enters again.

It has been reported that immunological memory can be classified as passive short-term memory or active long-term memory.

Q4. Vaccine and how it works? 3 Marks

A vaccine is a biological preparation that improves human immunity to a particular disease. A vaccine basically contains an agent that resembles a disease-causing microorganism, and is often made from weakened or killed forms of the microorganism or its toxins.

Upon injecting these vaccines in humans, the vaccine molecules stimulate the body's immune system to recognize the agent as foreign, destroy it, and memorize it, so that the immune system can more effortlessly familiarize with it and destroy any of these microorganisms that it later encounters.

Q5. Importance of Sperm mediated transgenesis? 5 Marks

The ability of sperm cells to carry exogenous DNA spontaneously into the oocyte during fertilization was first reported by Brackett and coworkers in 1971. However, the concept of sperm-mediated transgenesis rested for 18 years until scientists reported the use of spermatozoa as DNA carriers to produce transgenic mice. The scientific community was challenged by an unsuccessful attempt to replicate the experiment met this work with skepticism. Transgene delivery by sperm cells was used to produce transgenic animals in a wide variety of species, including cattle, pigs, rabbits, frogs, and zebra fish. Sperm-mediated transgenesis is the most straightforward technique to produce transgenic animals. The spermatozoa are incubated with the DNA containing the gene of interest followed by in vivo or in vitro insemination. DNA binds to the sperm's plasma membrane through specific DNA binding proteins. After internalization, they are carried into the oocyte upon fertilization.

The most appealing characteristics of using sperm as vectors to produce transgenic animals are its simplicity (no embryo manipulation is required) and the possibility of performing mass production of genetically modified animals through in vivo or in vitro insemination of many oocytes.

Q6. Role of nanotechnology in vaccine? 5 Marks

Nano vaccine

Nano is very vast field and it can be applied to any area, one of such area is vaccine. It provides a different route of administration of vaccine. Nano vaccine can be designed, manufactured and introduced into the human body to improve health, including cellular repairs at the molecular level. The nanomaterial is so small that it can easily enter the cell; therefore, nanomaterials can be used in vivo or in vitro for biological applications.

Following are the different way to deliver vaccine at nano level:

Nanobead: Inert solid bead Size range 20-200nm.

When antigens are adsorbed on the surface of bead it has been shown to stimulate CD8 -T cell response. The size of the bead plays a major role in eliciting a combined response of humoral and cell- mediated immunity. Antigen covalently linked to inert nano-beads with a size of ~50 nm is preferentially taken up by dendritic cells, thus inducing humoral as well as cell-mediated immune responses.

Polymeric nanoparticles: Biodegradable, biocompatible polymers have been approved for use in humans.

- Poly(D,L-lactide-co-glycolide) (PLG) and polylactide (PLA).

Q7. Write down the function of B-Lymphocyte? 2 Marks

The B lymphocytes and antibody cells detect pathogens when the antibodies on their surface attach to a specific foreign antigen. This antigen–antibody complex is taken up by the B cell and processed by proteolysis into peptides.

Q8. Benefits Zift over IVF? 2 Marks

Only benefit ZIFT over IVF is for women who are thought to have compromised embryo quality due to embryo in vitro culture. Placing zygotes back into their own natural incubators is thought to enhance development and improve pregnancy.

Q9. Disadvantages of Retrovirus mediated genes transfer for producing transgenic animals? 3 Marks

The disadvantages of this technique are the possible interference of viral regulatory elements with the expression of genes and the susceptibility of virus to de novo methylation resulting in silencing of the genes. Though viruses are highly adapted to the process of gene transfer however, using virus as the vector may also produce health hazards as they may potentially arouse cancer/leukemia.

Q10. What are the systems involved in cytometer? 3 Marks

Fluidics System: Flow cytometers use the principle of hydrodynamic focusing for presenting cells one at a time to a light source. The fluidics system consists of a central channel through which the sample is injected, enclosed by an outer sheath that contains faster flowing fluid.

Q11. How artificial skin made? Write the process in Points. 5 Marks

Skin is usually donated by other donors. Fibroblasts are removed from the donated skin and are frozen until they are needed. The fibroblasts are placed on a polymeric mesh scaffolding, gather oxygen, and grow new cells. The cells are then transferred to a culture system. After 4 weeks the polymer mesh dissolves and leaves behind a new layer of dermal skin. When the growth cycle is completed, they add more nutrients. Keratinocytes are added to the collagen and are exposed to air to form epidermal layers. The skin is now completed and is stored in sterile contains until ready to use.

Q12. What is Biopharmaceutical? 2 Marks

Biopharmaceuticals are defined as pharmaceuticals manufactured by biotechnology methods, with the products having biological sources, usually involving live organisms or their active components. A protein or nucleic acid used as a drug, which has been produced by living organisms as a result of biotechnology.

Q13. Application of erythropoietin? 3 Marks

- EPO can be used as a therapeutic agent to treat various disease conditions
- It can be used in the treatment of anemia resulting from chronic kidney disease and
- myelodysplasia, and in the treatment of cancer.
- There are various types of EPO produced for patients who are suffering from anemia.
- These are epoetin, darbepoetin, epoetin delta, PD poetin, and methoxy polyethylene glycol- epoetin beta. The list of erythropoietin produced by using r-DNA technology.

Q14. Write a note on Human recombinant insulin? 5 Marks

Insulin is a peptide hormone consisting of 51 amino acids. It is secreted by β cells of islets of Langerhans of pancreas. The hormone is responsible for maintaining normal blood glucose level in blood. Insulin is stored in the form of proinsulin which contains two polypeptide chains, A and B, and is connected with a third peptide C chain, which before secretion is cleaved with production of insulin and C-chain.The cleavage results in the removal of C-chain, and the A (21 amino acids) and B chain (30 amino acids) are linked by disulfide linkage to form mature insulin.

In the beginning, efforts were made to isolate mRNA for pre- and proinsulin from rat islets of Langerhans of pancreas and to synthesize cDNA. Thereafter, it was inserted into a plasmid. The recombinant plasmids were transferred into the E. coli cells, which secreted proinsulin. Scientists have chemically synthesized DNA sequences for two chains, A and B, of insulin and separately inserted into two pBR322 plasmids by the side of β-galactosidase gene. The recombinant plasmids were separately transferred into E. coli cells which secreted fused β galactosidase-A chain and β-galactosidase-B chain separately. These chains were isolated in pure form by detaching from βgalactosidase with yields of about 10 mg/24 g of healthy and transformed cells.

Q15. Name the vaccine Adjuvants?3 Marks

- MF59
- Monophosphoryl lipid A
- Montanide™
- Calcium phosphate nanoparticles.
- Immunostimulating complexes
- Cholesterol-bearing hydrophobized pullulan
- Nanoparticles (CHP)

Q16. Passive Immunity? 2 Mark

Development of immunity between the mother and child is called passive immunity because the fetus does not make any antibodies on its own, it only borrows them from the mother. It has been reported that passive immunity is only for a short duration of time, up to several months.

Q17. Classification of Artificial organ? 3 Marks

Artificial organs can conveniently be classed into four groups:

I. Bone/Joint Replacements (e.g. hip, knee, finger, total limb),

II. Skin/Soft Tissue Replacements (e.g. skin, breast, muscle),

III. Internal Organs (e.g. heart, kidney, blood vessels, liver, pancreas)

IV. Sensory Organs (e.g. eye, ear).

Q18. Procedure involves in ICSI? 3 Marks

The ovum is injected with single sperm in vitro with the help of micromanipulation devices. After the procedure the oocyte is placed into cell culture and checked for signs of fertilization. Once fertilization is accomplished, cell division occur and embryo is transferred into women's uterus.

Q19. Write a note on Adjuvant. 5 Marks

This substance used in combination with a specific antigen that produced a more robust immunity than the antigen alone.

1. It improves the immune response to vaccine antigen by different way:
1. Increasing immunogenicity of weak antigen.
2. Enhancing the speed and duration of the immune response.
3. Modulating antibody avidity, specificity, isotope or subclass distribution.

Examples:

- MF59
- Monophosphoryl lipid A
- Montanide™
- Calcium phosphate nanoparticles.
- Immunostimulating complexes
- Cholesterol-bearing hydrophobized pullulan
- Nanoparticles (CHP)

Q21. Describe in detail the procedure of artificial preparation of human organ? 5 Marks

Manufacturing Processes of Artificial Organs

The various processes involved in the manufacturing of artificial organs involve the following:

- **Tissue Engineering:** This is the use of a combination of cells, engineering and materials methods, and suitable biochemical and physico-chemical factors to improve or replace biological functions.
- **Bioprinting or 3D printing of tissues:** Bioprinting, where living cells are precisely printed in a certain pattern, has great potential and promise for fabricating engineered living organs.
- **Industrially by using biomaterials** which includes metals, ceramics, polymers and composites.
- **In the laboratory** by using stem cells under Regenerative Medicine

Q22. DNA vaccine? 3 Marks

DNA vaccination is a technique for protecting an organism against disease by injecting it with genetically engineered DNA to produce an immunological response. DNA vaccines consist of plasmids engineered to contain a protein encoding gene and all the necessary requisites.

Animals can be immunized against disease merely by injecting plasmids suspended in saline (salt) solution. And delivery of the plasmids to target cells is relatively easy: researchers identified five effective ways for delivering influenza virus plasmids to the animal body: by nasal spray; by injection into the muscle, vein, and under the skin; and by a so-called gene gun.

Q23. Nanopore Sequencing? 3 Marks

- The fourth-generation DNA sequencing technology
- Studies the interaction between DNA and protein, as well as between protein and protein.
- Have the potential to quickly and reliably sequence the entire human genome for less than $1000, and possibly for even less than $100.

- The detection principle is based on monitoring the ionic current passing through the nanopore as a voltage is applied across the membrane.
- When the nanopore is of molecular dimensions, passage of molecules (e.g., DNA) cause interruptions of the current level, leading to a signal.

Q24. Immunological memory? 5 Marks

When B cells and T cells are activated and begin to reproduce, some of their offspring can become long-lived memory cells. This process is called immunological memory. Memory cells

can remember each and every specific pathogen encountered and can enhance a strong response if the pathogen is detected or enters again. This kind of immunological response is basically adaptive in nature, because it occurs during the lifetime of an individual. It has been reported that immunological memory can be classified as passive short-term memory or active long-term memory.

Q25. Drug Abuse? 5 marks

A form of substance-related disorder. Patterned use of a drug in which the user consumes the substance in amounts or with methods which are harmful to themselves or others

Drugs most often associated with this term include: Alcohol, substituted amphetamines, barbiturates, benzodiazepines, cocaine, methaqualone, and opioids.

Q26. Trade Name of tissue Plasminogen activator? 3 Marks

There are various commercially available rtPAs in the market, which include Alteplase, Reteplase, and Tenecteplase.

Q27. Pharmacogenomics? 2 Marks

Pharmacogenomics: is the science of how genes affect the way people respond to drugs

- How genes affect, the way our body processes drugs (pharmacokinetics)
- The interaction of drugs with receptors (pharmacodynamics)
- The treatment efficacy and adverse side effects

Q28. Write a note on applications of DNA fingerprinting. 5 Marks

Diagnosis of Inherited Disorders

o Helps diagnose disorders in both prenatal and newborn babies

o Disorders may include cystic fibrosis, hemophilia, Huntington's disease, familial Alzheimer's, sickle cell anemia, thalassemia, and much more.

Crime

- Forensic science is the use of scientific knowledge in legal situations.
- The DNA profile of each individual is highly specific.
- The chances of two people having exactly the same DNA profile is 30,000 million to 1 (except for identical twins).

Personal Identification: The U.S. armed services are just beginning a program where they collect DNA fingerprints from all personnel for later use, in case they need to identify casualties or missing people.

Some Other Uses of DNA fingerprinting

- Identification of carcass of tissues
- Detection of somatic mutations or cancer
- Pathogen identification
- Detection of loci controlling quantitative traits or disease resistance
- Sex determination
- Individual identification

Q29. Differentiate between Allogeneic, xenogeneic and Isogenic cells? 3 Marks

Allogeneic cells come from the body of a donor of the same species; for example, the cells from one person (donor) can be used by another person (patient).

The cells which are isolated from two different species are called as **xenogeneic cells;** for example, the cells from one species (sheep) can be used in another species (human).

Isogenic cells are type of cell isolates from genetically identical individuals, such as twins, clones, or extremely inbred research animal models.

Q30. Define autologous transplantation and its importance? 3 Marks

The autologous cell transplantation is basically a transplantation in which cells obtained from patients can be transplanted into the same patient. One significance of the autologous

transplantation is that there is a high rate of success as the transplanted cells easily can be accepted by the patient's body and there are few problems associated with ejection and pathogen transmission.

Q31. Disadvantages of Live attenuated vaccine? 3 Marks

1. Storage conditions are to be strictly maintained.
2. Reversion is one of an important risk.
3. Vaccines can cause serious complications in immunocompromised patients

Q32. Classification of Dialysis Membrane? 3 Marks

- Dialysis membrane can be classified in to three based up on their polymeric composition and preparation process.
- Regenerated cellulosic membrane
- Modified cellulosic membrane
- Synthetic membrane

Q33. Note on Tissue Plasminogen Activator? 5 Marks

Tissue plasminogen activator (tPA) is another kind of serine protease protein that is involved in the breakdown of blood clots in humans. In the human body, tissue plasminogen is found on endothelial cells, the cells that line the blood vessels. As an enzyme, tPA catalyzes the conversion of plasminogen to plasmin, the major enzyme responsible for clot breakdown. It has been reported that tPA is used as clinical medicine to treat only embolic or thrombotic stroke.

Patients who have a problem related to blood clotting need a medicine to clot the blood immediately to avoid further releases of the blood from the body and tPA is considered to be an important medicine to promote blood clot. The tPA can be manufactured by using r-DNA technology and the production of tPA created this way is referred to as recombinant tissue plasminogen activator (rtPA). The basic principle behind blood clotting is that tPA catalyzes the conversion of plasminogen into plasmin, which is done by cleaving the single-chain plasminogen into two chains. These two chains are connected by a disulfide bond and the resulting molecule is called plasmin. In the patients, an increased enzymatic activity can cause hyper fibrinolysis, which demonstrates as an excessive bleeding.

Q34. What criteria new drugs must meet? 5 Marks

Drugs must address a new need or provide a significant “added benefit” over an existing medicine. Drugs must also meet these criteria:

- Must be safe
- Effective
- Of high quality
- Cost effective